# Durethan® BTC77ZH3.0EF

PA6-MX77 Envalior

Injection Molding, 77% Mineral Reinforced, Heat Stabilized, Thermal conductive material

ISO 1043 PA6-MX77

| Rheological properties | dry / cond | Unit | Test Standard |
|---|---|---|---|
| **ISO Data** | | | |
| Molding shrinkage, parallel | 0.7 / * | % | ISO 294-4, 2577 |
| Molding shrinkage, normal | 0.6 / * | % | ISO 294-4, 2577 |

| Mechanical Properties | dry / cond | Unit | Test Standard |
|---|---|---|---|
| **ISO Data** | | | |
| Tensile Modulus | 13600 / 5000 | MPa | ISO 527 |
| Stress at Break | 95 / 60 | MPa | ISO 527 |
| Strain at Break | 2.8 / 5 | % | ISO 527 |
| Impact Strength (Charpy), +23°C | 35 / 50 | kJ/m² | ISO 179/1eU |
| Impact Strength (Charpy), -30°C | 20 / 20 | kJ/m² | ISO 179/1eU |

| Thermal Properties | dry / cond | Unit | Test Standard |
|---|---|---|---|
| **ISO Data** | | | |
| Melting Temperature (10°C/min) | 222 / * | °C | ISO 11357-1/-3 |
| Temp. of deflection under load (1.80 MPa) | 142 / * | °C | ISO 75-1/-2 |
| Temp. of deflection under load (0.45 MPa) | 191 / * | °C | ISO 75-1/-2 |
| Coeff. of Linear Therm. Expansion, parallel | 40 / * | E-6/K | ISO 11359-1/-2 |
| Coeff. of Linear Therm. Expansion, normal | 40 / * | E-6/K | ISO 11359-1/-2 |

| Electrical Properties | dry / cond | Unit | Test Standard |
|---|---|---|---|
| **ISO Data** | | | |
| Volume Resistivity | >1E13 / - | Ohm*m | IEC 62631-3-1 |
| Electric Strength | 31 / - | kV/mm | IEC 60243-1 |
| Comparative tracking index | 600 / - | - | IEC 60112 |

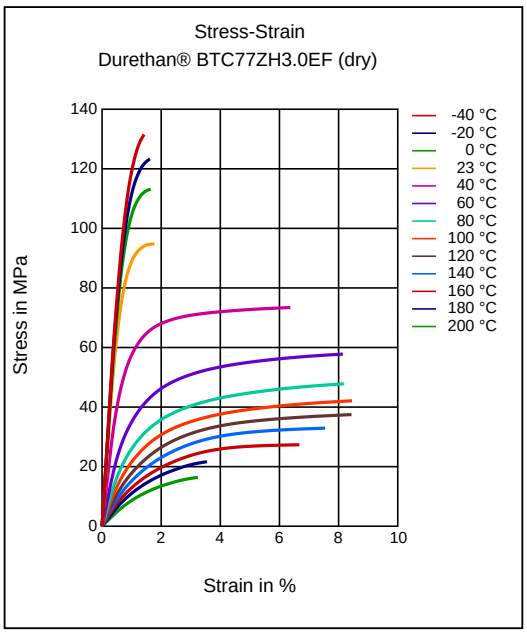
| Other Properties | dry / cond | Unit | Test Standard |
|---|---|---|---|
| **ISO Data** | | | |
| Density | 2510 / - | kg/m³ | ISO 1183 |

| Test specimen production | Value | Unit | Test Standard |
|---|---|---|---|
| **ISO Data** | | | |
| Injection Molding, melt temperature | 290 | °C | ISO 294 |
| Injection Molding, mold temperature | 80 | °C | ISO 294 |

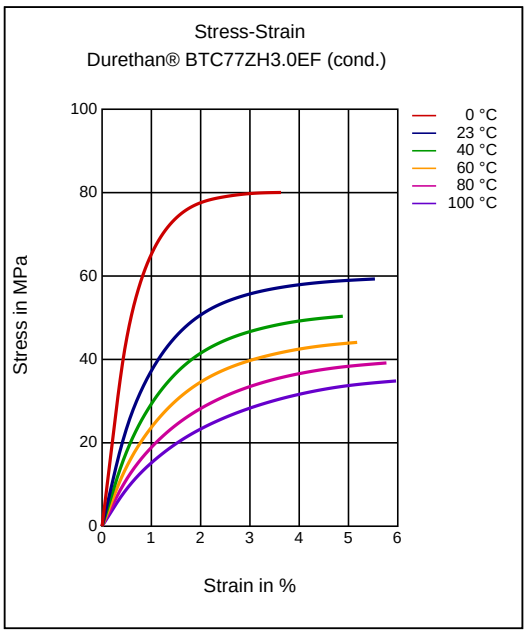
| Processing Recommendation Injection Molding | Value | Unit | Test Standard |
|---|---|---|---|
| Pre-drying - Temperature | 80 | °C | - |
| Pre-drying - Time | 2 - 6 | h | - |
| Melt temperature | 280 - 300 | °C | - |
| Mold temperature | 80 - 120 | °C | - |

## Diagrams

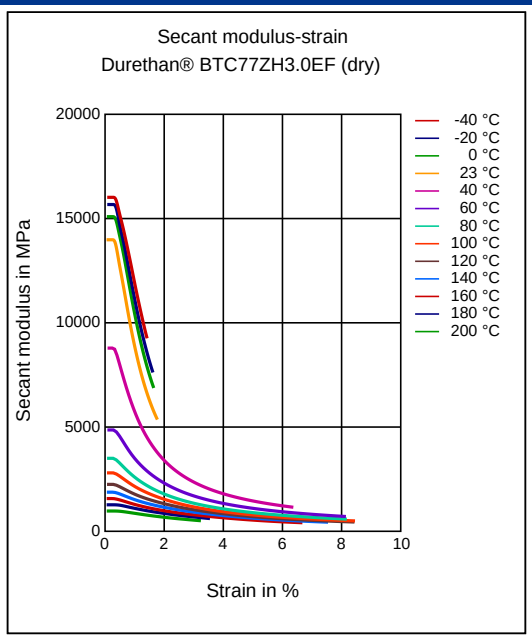
### Stress-strain



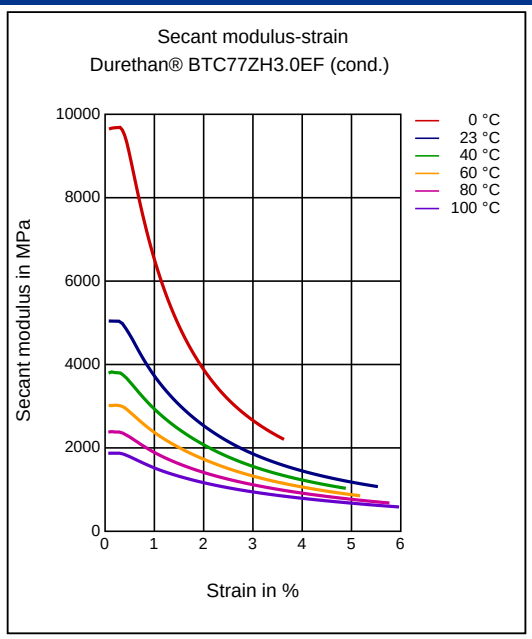

### Stress-strain




### Secant modulus-strain




### Secant modulus-strain




## Characteristics

### Processing
Injection Molding

### Special Characteristics
Heat aging stabilized

### Delivery form
Pellets

### Features
Thermally Conductivity

### Injection Molding
PREPROCESSING

Drying temperature dry air dryer: 80 °C

Drying time dry air dryer 2 - 6 h

PROCESSING
Melt temperature (Tmin - Tmax): 280 - 300 °C

Mold temperature: 80 - 120 °C

## Disclaimer

**Liability Exclusion**
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